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SYSTEM AND METHOD FOR OBTAINING PROFILE OF EYE LENS CAPSULE

Abstract

A system includes a controller with at least one processor and at least one non-transitory, tangible memory on which instructions are recorded for executing a method for obtaining a profile of a lens capsule of an eye. The profile is represented by respective central surfaces and respective equatorial surfaces separated at respective transition points. The controller is configured to obtain imaging data for a portion of the lens capsule visible through a pupil of the eye. The imaging data is transformed to an adjusted frame of reference and fitted to the respective central surfaces in a predefined central region of the lens capsule. The profile is obtained based on a set of fitting parameters for the respective central and equatorial surfaces. The respective central surfaces and respective equatorial surfaces may be represented as elliptical cones and skewed parabolas, respectively.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to a system and method for obtaining a profile of a lens capsule of an eye.

BACKGROUND

[0002] Humans have five basic senses: sight, hearing, smell, taste, and touch. Sight gives us the ability to visualize the world around us and connects us to our surroundings. Many people worldwide have issues with quality of vision and require the use of ophthalmic lenses, such as for example, intraocular lenses. The intraocular lens may be implanted into the eye in a cataract procedure to replace a human lens that has become cloudy. Having a profile of the lens capsule of the eye, prior to the procedure, assists in the selection of the intraocular lens.

SUMMARY

[0003] Disclosed herein is a system having a controller with at least one processor and at least one non-transitory, tangible memory on which instructions are recorded for executing a method for obtaining a profile of a lens capsule of an eye. Execution of the instructions by the processor causes the controller to obtain imaging data for a portion of the lens capsule visible through a pupil of the eye. The imaging data includes posterior datapoints and anterior datapoints and is transformed to an adjusted frame of reference having a first axis (X) and a second axis (Y). Also disclosed is a corresponding method for obtaining a profile of a lens capsule of an eye

[0004] The profile is represented by respective central surfaces and respective equatorial surfaces separated by respective transition points. The controller is configured to fit the imaging data in the adjusted frame of reference to the respective central surfaces in a predefined central region of the lens capsule. The method includes obtaining a transition coordinate as a coordinate value of the respective transition points in a positive X-domain. The controller is configured to determine a set of fitting parameters for the respective central surfaces and the respective equatorial surfaces based on the transition coordinate and a plurality of constraints. The profile is obtained based on the set of fitting parameters for the respective central surfaces and the respective equatorial surfaces.

[0005] The controller may be configured to select an intraocular lens based at least partially on the profile of the lens capsule. Transforming the imaging data to the adjusted frame of reference includes fitting the posterior datapoints and the anterior datapoints to a first circle and a second circle, respectively, and determining intersection points of the first circle and the second circle. Transforming the imaging data to the adjusted frame of reference includes transforming the posterior datapoints and the anterior datapoints such that they are centered about a respective center of the intersection points and rotated such that a tilt angle of rotation is zero.

[0006] The controller is configured to fit the respective central surfaces in the adjusted frame of

reference to respective conic equations for a conic surface. The controller is configured to determine a conic x-intercept as a coordinate on the first axis (X) where the respective conic equations intersect in a positive-x domain, the transition coordinate being a product of the conic x-intercept and a predefined constant, the predefined constant being less than 1.

[0007] The respective central surfaces include a central anterior surface. The controller may be configured to represent the central anterior surface as an elliptical cone characterized by a first plurality of variables (Ka, Qa, Ra), the first plurality of variables (Ka, Qa, Ra) being obtained by fitting the imaging data to the central anterior surface in the predefined central region in the adjusted frame of reference. The central anterior surface Ca(x) may be defined as:

$$[00001]Ca(x) = Ka - \frac{\sqrt{-(1+Qa)[(1+Qa)^2x^2 - Ra^2]}}{(1+Qa)}.$$

[0008] The respective central surfaces include a central posterior surface. The controller may be configured to represent the central posterior surface as an elliptical cone characterized by a second plurality of variables (Kp, Qp, Rp), the second plurality of parameters (Kp, Qp, Rp) being obtained by fitting the imaging data to the central posterior surface in the predefined central region in the adjusted frame of reference. The central posterior surface Cp(x) may be defined as:

$$[00002]Cp(x) = Kp + \frac{\sqrt{-(1+Qp)[(1+Qp)^2x^2 - Rp^2]}}{(1+Qp)}.$$

[0009] The set of fitting parameters include a first anterior parameter (Ga), a second anterior parameter (Pa), a first posterior parameter (Gp), a second posterior parameter (Pp) and respective coordinates (Xe, Ye) of a vertex in the adjusted frame of reference. The respective equatorial surfaces include an equatorial anterior surface and an equatorial posterior surface meeting at the vertex. The equatorial anterior surface and the equatorial posterior surface are represented by respective skewed parabola functions. The equatorial anterior surface is based in part on the first anterior parameter (Ga), the second anterior parameter (Pa) and the respective coordinates (Xe, Ye) of the vertex. The equatorial anterior surface Ea(x) may be defined as: $Ea(x) = -[(1-Ga(Xe-x))] [2\sqrt{\text{square root over } (Pa(Xe-x))} - Ye]$. The equatorial posterior surface is based in part on the first posterior parameter (Gp), the second posterior parameter (Pp) and the respective coordinates (Xe, Ye) of the vertex. The equatorial posterior surface Ep(x) may be defined as: $Ep(x) = +[(1-Gp(Xe-x))] [2\sqrt{\text{square root over } (Pp(Xe-x))} + Ye]$.

[0010] The respective central surfaces include a central posterior surface and a central anterior surface. The respective equatorial surfaces include an equatorial anterior surface and an equatorial posterior surface. The plurality of constraints includes a first equation matching respective values of the central anterior surface and the equatorial anterior surface at the transition coordinate, and a second equation matching the respective values of the central posterior surface and the equatorial posterior surface at the transition coordinate.

[0011] The plurality of constraints includes a third equation matching respective first derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate, and a fourth equation matching the respective first derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate. The plurality of constraints includes a fifth equation matching respective second derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate, and a sixth equation matching the respective second derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate.

[0012] Disclosed herein is a system including a controller having at least one processor and at least one non-transitory, tangible memory on which instructions are recorded for executing a method for obtaining a profile of a lens capsule of an eye. The profile is represented by respective central surfaces and respective equatorial surfaces separated by respective transition points. Execution of the instructions by the processor causes the controller to obtain a lens diameter and at least two variables from a set of variables, the set of variables including a lens thickness, a central anterior apex and a central posterior apex. A transition coordinate is set as a product of the lens diameter

and a predefined constant, the predefined constant being less than 0.5.

[0013] The controller is configured to obtain a first plurality of variables (Ka, Qa, Ra) and a pair of anterior parameters (Ga, Pa) by simultaneously solving a first group of constraints based in part on the transition coordinate. The controller is configured to obtain a second plurality of variables (Kp, Qp, Rp) and a pair of posterior parameters (Gp, Pp) by simultaneously solving a second group of constraints based in part on the transition coordinate.

[0014] The controller is configured to obtain the profile based on the first plurality of variables (Ka, Qa, Ra), the pair of anterior parameters (Ga, Pa), the second plurality of variables (Kp, Qp, Rp) and the pair of posterior parameter (Gp, Pp). An updated value of the lens diameter is obtained based on the profile, and an updated value of the transition coordinate is obtained based on the updated value of the lens diameter.

[0015] When a difference between the updated value of the transition coordinate and the transition coordinate is greater than a predefined threshold, the controller is configured to update the first plurality of variables (Ka, Qa, Ra) and the pair of anterior parameters (Ga, Pa) by simultaneously solving the first group of constraints based in part on the updated value of the transition coordinate. When the difference between the updated value of the transition coordinate and the transition coordinate is greater than the predefined threshold, the controller is configured to update the second plurality of variables (Kp, Qp, Rp) and the pair of posterior parameter (Gp, Pp) by simultaneously solving the second group of constraints based in part on the updated value of the transition coordinate.

[0016] The respective central surfaces include a central anterior surface and a central posterior surface, the respective equatorial surfaces including an equatorial anterior surface and an equatorial posterior surface. The controller is configured to represent the central anterior surface and the central posterior surface as respective elliptical cones characterized by the first plurality of variables (Ka, Qa, Ra) and the second plurality of variables (Kp, Qp, Rp), respectively. The controller is configured to represent the equatorial anterior surface and the equatorial posterior surface as respective skewed parabolas characterized by the pair of anterior parameters (Ga, Pa) and the pair of posterior parameters (Gp, Pp), respectively.

[0017] The first group of constraints includes: a first equation matching respective values of the central anterior surface and the equatorial anterior surface at the transition coordinate; a second equation matching respective first derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate; a third equation matching respective second derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate; and a fourth equation matching a respective coordinate of the central anterior surface to the central anterior apex.

[0018] The second group of constraints includes: a fifth equation matching the respective values of the central posterior surface and the equatorial posterior surface at the transition coordinate; a sixth equation matching respective first derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate; a seventh equation matching respective second derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate; and an eighth equation matching a respective coordinate of the central posterior surface to the central posterior apex.

[0019] The above features and advantages and other features and advantages of the present disclosure are readily apparent from the following detailed description of the best modes for carrying out the disclosure when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. **1** is a schematic illustration of a system for obtaining a profile of a lens capsule of an eye, the system having a controller;

[0021] FIG. **2** is a schematic flowchart for a method executable by the controller of FIG. **1**, in accordance with a first embodiment;

[0022] FIG. **3** is a schematic fragmentary example of a sectional image of an eye, the image having posterior datapoints and anterior datapoints of a lens capsule;

[0023] FIG. **4** is a schematic illustration of the posterior datapoints and anterior datapoints of FIG. **3**, in an original frame of reference;

[0024] FIG. **5** is a schematic illustration of the posterior datapoints and anterior datapoints of FIG. **4**, after transformation to an adjusted frame of reference;

[0025] FIG. **6** is a schematic diagram of a profile of the lens capsule obtained by the system of FIG. **1**, in the adjusted frame of reference;

[0026] FIG. **7** is a schematic diagram of the profile of the lens capsule shown in FIG. **6**, in the original frame of reference; and

[0027] FIG. **8** is a schematic flowchart for a method executable by the controller of FIG. **1**, in accordance with a second embodiment.

DETAILED DESCRIPTION

[0028] Referring to the drawings, wherein like reference numbers refer to like components, FIG. **1** schematically illustrates system **10** for obtaining a profile of a lens capsule of an eye. Referring to FIG. **1**, the system **10** includes a controller **C** having at least one processor **12** and at least one memory **14** (or non-transitory, tangible computer readable storage medium) on which are recorded instructions for executing one or more methods. Method **100** and Method **300** are shown in and described below with reference to FIGS. **2** and **8**, respectively.

[0029] Referring to FIG. **1**, the system **10** may include a user interface **16** for collecting user data from one or more clinical facilities or electronic medical record units. The system **10** may include a data management unit **18** for storing and/or facilitating transfer of the user data and other functions. The various components of the system **10** may be configured to communicate via a short-range network **20** and/or a long-range network **22**. Referring to FIG. **1**, the controller **C** may be in communication with a remote server **24** and/or a cloud unit **26**, which may include one or more servers hosted on the Internet to store, manage, and process data. The cloud unit **26** may be a private or public source of information maintained by an organization, such as for example, a research institute, a company, a university and/or a hospital.

[0030] Referring to FIG. **1**, the short-range network **20** may be a bus implemented in various ways, such as for example, a serial communication bus in the form of a local area network. The local area network may include, but is not limited to, a Controller Area Network (CAN), a Controller Area Network with Flexible Data Rate (CAN-FD), Ethernet, blue tooth, WIFI and other forms of data connection. The long-range network **22** may be a Wireless Local Area Network (LAN) which links multiple devices using a wireless distribution method, a Wireless Metropolitan Area Networks (MAN) which connects several wireless LANs or a Wireless Wide Area Network (WAN) which covers large areas such as neighboring towns and cities. Other types of connections may be employed.

[0031] The controller **C** may be configured to receive and transmit wireless communication to the remote server **24** through a mobile application **28**, shown in FIG. **1**. The mobile application **28** may be in communication with the controller **C** via the short-range network **20** such that it has access to the data in the controller **C**. In one example, the mobile application **28** is physically connected (e.g. wired) to the controller **C**. In another example, the mobile application **28** is embedded in the controller **C**. The circuitry and components of a remote server **24** and mobile application **28** (“apps”) available to those skilled in the art may be employed.

[0032] Referring to FIG. 1, the user interface **16** and/or the controller C may be configured to communicate with an imaging device **30**, which may be an optical coherence tomography machine. The imaging device **30** may be an ultrasound machine, a magnetic resonance imaging machine or other imaging device available to those skilled in the art. Additionally, the user interface **16** and/or the controller C may be in communication with a profile output module **32** and a lens selection module **34** for selecting an intraocular lens **36**, as described below.

[0033] Referring to FIG. 3, an example image of an eye E is shown. As described below and referring to FIG. 3, the controller C is configured to obtain imaging data for a portion **140** of a lens capsule **142** visible through the pupil **144** of the eye E. The lens capsule **142** has a lens thickness **146** and a lens diameter **148**. Also shown in FIG. 3 are the cornea **152** and iris **154**. It is understood that FIG. 3 is not to scale.

[0034] Referring now to FIG. 2, a flowchart of method **100** is shown. Method **100** need not be applied in the specific order recited herein and some blocks may be omitted. Per block **102** of FIG. 2, the method **100** includes obtaining imaging data for a portion **140** of the lens capsule **142** visible through the pupil **144** of the eye E. Method **100** enables a prediction of the entire shape of the lens capsule **142**, including the portions of the shape that are obscured by the iris **154**, using the portions **140** of the surface which is visible through the open pupil **144**. The imaging data may be obtained via ultrasound bio-microscopy, optical coherence tomography, magnetic resonance imaging or any other imaging modality available to those skilled in the art. The imaging data may be derived from a single image or from multiple images. The imaging data may be obtained from imaging device **30**.

[0035] Referring to FIG. 3, the imaging data includes posterior datapoints **160** (at a posterior side P) and anterior datapoints **170** (at an anterior side A). The posterior datapoints **160** and anterior datapoints **170** are shown plotted in FIG. 4 in pixel units (referred to herein as the original frame of reference **150**), with a horizontal axis Q and a vertical axis R. The image shown in FIG. 4 (with posterior side P being above the anterior side A) is flipped vertically compared to FIG. 3 (with anterior side A being above the posterior side P) since the coordinate value in the vertical axis R advances going downwards in FIG. 3.

[0036] Per block **104** of FIG. 2, the method **100** includes transforming the posterior datapoints **160** and anterior datapoints **170** from the original frame of reference **150** (shown in FIG. 4) to an adjusted frame of reference **200** (shown in FIGS. 5 and 6). First, referring to FIG. 4, the posterior datapoints **160** are fitted to the equation for a first circle **162** at the posterior side P. The anterior datapoints **170** are fitted to the equation for a second circle **172** at the anterior side A. The first circle **162** and the second circle **172** intersect at intersection points **180**, **182**, shown in FIG. 4.

[0037] Second, referring to FIG. 4, the controller C is configured to determine a center **184** between the intersection points **180**, **182**. The coordinate-transformed points (in the adjusted frame of reference **200** of FIG. 5) are obtained by centering the posterior datapoints **160** and the anterior datapoints **170** about the center **184** and rotating the centered coordinates by an angle exactly opposite to a tilt angle **188**, such that the resulting tilt angle **188** becomes zero. The tilt angle **188** may be obtained as an arctangent of a ratio of the increase in the respective vertical coordinate value divided by an increase in the respective horizontal coordinate value between the intersection points **180**, **182**. The adjusted frame of reference **200** is shown in FIGS. 5-6 and has an X axis and a Y axis.

[0038] Per block **106** of FIG. 2 and referring to FIG. 5, the controller C is configured to fit the posterior datapoints **160** and the anterior datapoints **170** in the adjusted frame of reference **200** to respective central surfaces **210** (see FIG. 5) in a predefined central region **205** of the lens capsule **142**. In one example, the predefined central region **205** is in a range 3-7 mm. In one example, the predefined central region **205** is about 5 mm, corresponding to an average-sized open pupil. The predefined central region **205** may be correlated to the opening diameter of the pupil **144** of a specific patient.

[0039] In the embodiment shown, the respective central surfaces **210** are elliptical cones. The respective central surfaces **210** may be other types of conic surfaces. It is understood that the form of the respective central surfaces **210** may be varied. Referring to FIG. 5, trace **202** and trace **212** show extrapolation of the respective central surfaces **210** at the posterior side P and the anterior side A, respectively. The respective central surfaces **210** are extrapolated to determine a positive conic intersection point **220** and a negative conic intersection point **222**. The controller is configured to determine a conic x-intercept **224** as a coordinate on the X axis where the respective conic equations intersect in a positive-x domain. In other words, the conic x-intercept **224** is the X-coordinate of the positive conic intersection point **220**.

[0040] Referring to FIGS. 6-7, the profile L is represented by respective central surfaces **210** and respective equatorial surfaces **240**. The respective equatorial surfaces **240** include equatorial posterior surfaces **252** and equatorial anterior surfaces **262**. The central anterior surface **260** is flanked by the equatorial anterior surface **262** on either side. The central posterior surface **250** is flanked by the equatorial posterior surface **252** on either side. Referring to FIG. 6, the profile L of the lens capsule **142** is symmetric about the X-axis at X=0.

[0041] Per block **108** of FIG. 2, the controller C is configured to obtain a transition coordinate X_t (see FIG. 6) which is the coordinate value of the respective transition points **274** in a positive X-domain. Stated differently, the transition coordinate X_t is the positive coordinate on the X axis for the respective transition points **274**. Referring to FIG. 6, the respective transition points **274** in the positive X-domain and the negative X-domain (corresponding to coordinate— X_t) are an equal distance from the X=0 line. The transition coordinate X_t is set as a product of the conic x-intercept **224** (X_i) and a predefined constant F, such that $X_t = F * X_i$. The predefined constant F is less than 1. In one example, the predefined constant F is within a range of 0.5 to 0.9. In one example, the predefined constant F is set to 0.7.

[0042] Referring to FIG. 6, the central posterior surface **250** and the central anterior surface **260** may be generated from expressions for the conic equations over the range $-X_t < x < X_t$, where X_t is the transition coordinate. In the example shown, the central anterior surface **260** is represented as an elliptical cone characterized by a first plurality of variables (K_a , Q_a , R_a). The central anterior surface **260** or $Ca(x)$ is defined as:

$$[00003] Ca(x) = K_a - \frac{\sqrt{-(1 + Q_a)[(1 + Q_a)^2 x^2 - R_a^2]}}{(1 + Q_a)}.$$

The first plurality of variables (K_a , Q_a , R_a) are obtained by fitting the anterior datapoints **170** (in the adjusted frame of reference **200**) in the predefined central region **205**.

[0043] The central posterior surface **250** is represented as an elliptical cone characterized by a second plurality of variables (K_p , Q_p , R_p). The central posterior surface **250** or $Cp(x)$ is defined as:

$$[00004] Cp(x) = K_p + \frac{\sqrt{-(1 + Q_p)[(1 + Q_p)^2 x^2 - R_p^2]}}{(1 + Q_p)}.$$

The second plurality of parameters (K_p , Q_p , R_p) being obtained by fitting the posterior datapoints **160** (in the adjusted frame of reference **200**) in the predefined central region **205**.

[0044] Referring to FIGS. 6-7, the equatorial posterior surfaces **252** and the equatorial anterior surfaces **262** may be represented by respective skewed parabola functions. The equatorial anterior surfaces **262** and the equatorial posterior surfaces **252** meet at a vertex **270** in the positive X-domain and another vertex **272** in the negative X-domain.

[0045] Per block **110** of FIG. 2, the method **100** includes determining a set of fitting parameters for the respective central surfaces **210** and the respective equatorial surfaces **240**. The set of fitting parameters include a first anterior parameter (G_a), a second anterior parameter (P_a), a first posterior parameter (G_p), a second posterior parameter (P_p) and respective coordinates (X_e , Y_e) of the vertex **270** in the positive X-domain in the adjusted frame of reference **200**.

[0046] The equatorial posterior surface **252** (see FIGS. 6 and 7) is based in part on the first posterior parameter (G_p), the second posterior parameter (P_p) and the respective coordinates (X_e , Y_e) of the vertex **270** in the positive X-domain. The equatorial posterior surface **252** or $Ep(x)$ is

defined as:

$$[00005]Ep(x) = +[(1 - Gp(Xe - x))][2\sqrt{Pp(Xe - x)} + Ye] .$$

The equatorial anterior surface **262** (see FIGS. **6** and **7**) based in part on the first anterior parameter (Ga), the second anterior parameter (Pa) and respective coordinates (Xe, Ye) of the vertex **270**. The equatorial anterior surface **262** or Ea(x) is defined as:

$$[00006]Ea(x) = -[(1 - Ga(Xe - x))][2\sqrt{Pa(Xe - x)} - Ye] .$$

[0047] The set of fitting parameters are based on the transition coordinate Xt and a plurality of constraints. The plurality of constraints includes first, second, third, fourth, fifth and sixth equations. In the example shown, there are six fitting parameters and six constraint equations. The six constraint equations may be solved numerically, for example, using the MATLAB function fsolve, employing the trust-region algorithm. Other numerical algorithms available to those skilled in the art may be employed.

[0048] The first equation matches respective values of the central anterior surface **260** and the equatorial anterior surface **262** at the transition coordinate Xt, as follows: Ca(Xt)=Ea(Xt). The second equation matches the respective values of the central posterior surface **250** and the equatorial posterior surface **252** at the transition coordinate Xt, as follows: Cp(Xt)=Ep(Xt).

[0049] The third equation matches respective first derivatives of the central anterior surface **260** and the equatorial anterior surface **262** at the transition coordinate Xt, as follows:

$$[00007]\frac{dCa(x)}{dx} \cdot \text{Math}_{x = Xt} = \frac{dEa(x)}{dx} \cdot \text{Math}_{x = Xt} .$$

The fourth equation matches the respective first derivatives of the central posterior surface **250** and the equatorial posterior surface **252** at the transition coordinate Xt, as follows:

$$[00008]\frac{dCp(x)}{dx} \cdot \text{Math}_{x = Xt} = \frac{dEp(x)}{dx} \cdot \text{Math}_{x = Xt} .$$

The fifth equation matches respective second derivatives of the central anterior surface **260** and the equatorial anterior surface **262** at the transition coordinate Xt, as follows:

$$[00009]\frac{d^2 Ca(x)}{dx^2} \cdot \text{Math}_{x = Xt} = \frac{d^2 Ea(x)}{dx^2} \cdot \text{Math}_{x = Xt} .$$

The sixth equation matches the respective second derivatives of the central posterior surface **250** and the equatorial posterior surface **252** at the transition coordinate Xt, as follows:

$$[00010]\frac{d^2 Cp(x)}{dx^2} \cdot \text{Math}_{x = Xt} = \frac{d^2 Ep(x)}{dx^2} \cdot \text{Math}_{x = Xt} .$$

[0050] Per block **112** of FIG. **2**, the controller C is configured to obtain the profile L based on the set of fitting parameters (obtained in block **110**) applied to the respective central surfaces **210** and the respective equatorial surfaces **240**. FIG. **6** shows the profile L after the set of fitting parameters have been obtained. Additionally, per block **112**, the profile L may be re-adjusted to the original frame of reference **150** shown in FIG. **4**. FIG. **7** is a schematic diagram of the profile L after the opposite transformation from that of block **104**, including de-centering and re-tilting. In order to provide visualization for clinicians, the profile L in the original frame of reference **150** may be superimposed onto the image of the eye E in FIG. **3**.

[0051] Also, per block **112**, the controller C may be configured to select an intraocular lens **36** based at least partially on the profile L of the lens capsule **142**. Obtaining an accurate shape of the lens capsule L optimizes selection of the power of the intraocular lens **36**. This effect is heightened where the intraocular lens **36** is an accommodative lens which may change its shape in response to external forces. In other words, the intraocular lens **36** may react differently to the same accommodative changes mediated by the ciliary muscles, depending on the geometric dimension and shape of the lens capsule **142**.

[0052] Referring now to FIG. **8**, a flowchart of the method **300**, executable by the controller C of FIG. **1**, is shown. Method **300** need not be applied in the specific order recited herein and some blocks may be omitted. Per block **302** of FIG. **8**, the controller C is configured to obtain a lens diameter **148** (see FIG. **3**) and at least two variables from a set of variables. The set of variables includes a lens thickness **146** (see FIG. **3**), respective coordinates on the Y-axis of the central anterior apex **264** (see FIG. **6**) and the central posterior apex **254** (see FIG. **6**). The set of variables

may be obtained from the imaging device **30** of FIG. **1** or any other source. Method **300** enables a prediction of the entire shape of the lens capsule **142** using a handful of parameters.

[0053] Referring to FIG. **7**, the method **300** includes representing the profile **L** with respective central surfaces **210** and respective equatorial surfaces **240** in the adjusted frame of reference **200**, as shown in FIG. **6**. The respective central surfaces **210** include a central anterior surface **260** and a central posterior surface **250**, represented as elliptical cones characterized by a first plurality of variables (Ka, Qa, Ra) and a second plurality of variables (Kp, Qp, Rp), respectively. The respective equatorial surfaces **240** include an equatorial anterior surface **262** and an equatorial posterior surface **252**, represented as skewed parabolas characterized by a pair of anterior parameters (Ga, Pa) and a pair of posterior parameters (Gp, Pp), respectively.

[0054] Per block **304** of FIG. **8**, the controller **C** is configured to set the transition coordinate X_t (see FIG. **6**) as a product of the lens diameter **148** (LD) and a predefined constant J , such that $X_t = J \cdot LD$. The predefined constant J is less than 0.5. In one example, the predefined constant J is 0.35. The transition coordinate X_t may also be obtained as a product of the respective X-coordinate (X_e) of the vertex **270** in the positive X-domain and the predefined constant J , such that $X_t = J \cdot (2 \cdot X_e)$. The lens diameter **148** (LD) may be set as twice the value of the respective X-coordinate (X_e) of the vertex **270**, $LD = 2 \cdot X_e$.

[0055] Per block **306** of FIG. **8**, the controller **C** is configured to obtain the first plurality of variables (Ka, Qa, Ra) and the pair of anterior parameters (Ga, Pa) by simultaneously solving a first group of constraints based in part on the transition coordinate X_t . The first group of constraints includes four equations which may be solved numerically, for example, using the MATLAB function *fsolve*. Since there are five unknowns and four equations, the Levenberg-Marquardt method may be used. Other numerical algorithms available to those skilled in the art may be employed.

[0056] The first equation matches respective values of the central anterior surface **260** and the equatorial anterior surface **262** at the transition coordinate X_t , as follows: $Ca(X_t) = Ea(X_t)$. The second equation matches respective first derivatives of the central anterior surface **260** and the equatorial anterior surface **262** at the transition coordinate X_t , as follows:

$$[00011] \frac{dCa(x)}{dx} \cdot \text{Math}_{x = X_t} = \frac{dEa(x)}{dx} \cdot \text{Math}_{x = X_t} .$$

The third equation matches respective second derivatives of the central anterior surface **260** and the equatorial anterior surface **262** at the transition coordinate X_t , as follows:

$$[00012] \frac{d^2 Ca(x)}{dx^2} \cdot \text{Math}_{x = X_t} = \frac{d^2 Ea(x)}{dx^2} \cdot \text{Math}_{x = X_t} .$$

The fourth equation matches the respective coordinates (Y coordinate) of the central anterior surface **260** and the central anterior apex **264** (Y_{ac}) when the X coordinate is zero, such that $Ca(0) = Y_{ac}$.

[0057] Per block **308** of FIG. **8**, the controller **C** is configured to obtain the second plurality of variables (Kp, Qp, Rp) and the pair of posterior parameter (Gp, Pp) by simultaneously solving a second group of constraints based in part on the transition coordinate X_t . The second group of constraints includes four equations (fifth through eighth equations) which may be solved numerically, for example, using the MATLAB function *fsolve*. Since there are five unknowns and four equations, the Levenberg-Marquardt method may be used. Other numerical algorithms available to those skilled in the art may be employed.

[0058] The fifth equation matches the respective values of the central posterior surface **250** and the equatorial posterior surface **252** at the transition coordinate X_t , as follows: $Cp(X_t) = Ep(X_t)$. The sixth equation matches the respective first derivatives of the central posterior surface **250** and the equatorial posterior surface **252** at the transition coordinate X_t , as follows:

$$[00013] \frac{dCp(x)}{dx} \cdot \text{Math}_{x = X_t} = \frac{dEp(x)}{dx} \cdot \text{Math}_{x = X_t} .$$

seventh equation matches the respective second derivatives of the central posterior surface **250** and the equatorial posterior surface **252** at the transition coordinate X_t , as follows:

$$[00014] \frac{d^2 C_p(x)}{dx^2} \cdot \text{Math.}_x = X_t = \frac{d^2 E_p(x)}{dx^2} \cdot \text{Math.}_x = X_t \cdot$$

The eighth equation matches the respective coordinates (Y coordinate) of the central posterior surface **250** and the central posterior apex **254** (Ypc) when the X coordinate is zero, such that Cp(0)=Ypc. The central anterior apex **264** (Yac), central posterior apex **254** (Ypc) and the lens thickness **146** are related as follows: [Ypc=Yac+Lens Thickness].

[0059] The output of block **306** may be used to obtain the central anterior surface **260** and the equatorial anterior surface **262**. The central anterior surface **260** or Ca(x) is defined as:

$$[00015] Ca(x) = Ka - \frac{\sqrt{-(1+Qa)[(1+Qa)^2 x^2 - Ra^2]}}{(1+Qa)}.$$

The equatorial anterior surface **262** or Ea(x) is defined as: Ea(x)=-[(1-Ga(Xe-x))][2 Pa(Xe-x)-Ye].

[0060] The output of block **308** may be used to obtain the central posterior surface **250** and the equatorial posterior surface **252**. The central posterior surface **250** or Cp(x) is defined as:

$$[00016] Cp(x) = Kp + \frac{\sqrt{-(1+Qp)[(1+Qp)^2 x^2 - Rp^2]}}{(1+Qp)}.$$

The equatorial posterior surface **252** or Ep(x) is defined as: Ep(x)=+[(1-Gp(Xe-x))][2 Pp(Xe-x)+Ye].

[0061] Per block **310** of FIG. **8**, the controller C is configured to obtain the profile L based on the first plurality of variables (Ka, Qa, Ra), the pair of anterior parameters (Ga, Pa), the second plurality of variables (Kp, Qp, Rp) and the pair of posterior parameter (Gp, Pp). The controller C is configured to obtain an updated value of the lens diameter **148** based on the profile L, and an updated value of the transition coordinate Xt based on the updated value of the lens diameter **148**.

[0062] Per block **312** of FIG. **8**, the controller C is configured to determine if a difference between the updated value of the transition coordinate and the transition coordinate is less than a predefined threshold, in other words, if the updated value of the transition coordinate and the transition coordinate converge to within the predefined threshold. If so, the method **300** is ended. The controller C may be configured to select an intraocular lens **36** based at least partially on the profile L of the lens capsule **142**.

[0063] If not, as shown by line **313**, the method **300** loops back to block **306** and the controller C is configured to update the first plurality of variables (Ka, Qa, Ra) and the pair of anterior parameters (Ga, Pa) by simultaneously solving the first group of constraints based in part on the updated value of the transition coordinate. Additionally, the controller C is configured to update the second plurality of variables (Kp, Qp, Rp) and the pair of posterior parameter (Gp, Pp) by simultaneously solving the second group of constraints based in part on the updated value of the transition coordinate.

[0064] In summary, the system **10** (via execution of the method **100** and/or method **300**) enables the prediction of a profile L of the lens capsule **142** with relatively high accuracy while requiring a relatively small number of parameters. The system **10** uses separate parameter values for the anterior side A and the posterior side P of the lens capsule **142**, thus capturing the physiologic asymmetric nature of the shape of the lens capsule **142**, which can be flatter on one side compared to the other.

[0065] The controller C of FIG. **1** includes a computer-readable medium (also referred to as a processor-readable medium), including a non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Non-volatile media may include, for example, optical or magnetic disks and other persistent memory. Volatile media may include, for example, dynamic random-access memory (DRAM), which may constitute a main memory. Such instructions may be transmitted by one or more transmission media, including coaxial cables, copper wire and fiber optics, including the wires that comprise a system bus coupled to a processor of a computer. Some forms of

computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, other magnetic medium, a CD-ROM, DVD, other optical medium, punch cards, paper tape, other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH-EEPROM, other memory chip or cartridge, or other medium from which a computer can read.

[0066] Look-up tables, databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), etc. Each such data store may be included within a computing device employing a computer operating system such as one of those mentioned above and may be accessed via a network in one or more of a variety of manners. A file system may be accessible from a computer operating system and may include files stored in various formats. An RDBMS may employ the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

[0067] The detailed description and the drawings or FIGS. are supportive and descriptive of the disclosure, but the scope of the disclosure is defined solely by the claims. While some of the best modes and other embodiments for carrying out the claimed disclosure have been described in detail, various alternative designs and embodiments exist for practicing the disclosure defined in the appended claims. Furthermore, the embodiments shown in the drawings or the characteristics of various embodiments mentioned in the present description are not necessarily to be understood as embodiments independent of each other. Rather, it is possible that each of the characteristics described in one of the examples of an embodiment can be combined with one or a plurality of other desired characteristics from other embodiments, resulting in other embodiments not described in words or by reference to the drawings. Accordingly, such other embodiments fall within the adjusted framework of the scope of the appended claims.

Claims

1. A system comprising: a controller having at least one processor and at least one non-transitory, tangible memory on which instructions are recorded for executing a method for obtaining a profile of a lens capsule of an eye; wherein the profile is represented by respective central surfaces and respective equatorial surfaces separated by respective transition points; wherein execution of the instructions by the processor causes the controller to: obtain imaging data for a portion of the lens capsule visible through a pupil of the eye, the imaging data including posterior datapoints and anterior datapoints; transform the imaging data to an adjusted frame of reference having a first axis (X) and a second axis (Y); fit the imaging data to the respective central surfaces in a predefined central region in the adjusted frame of reference; obtain a transition coordinate as a coordinate value of the respective transition points in a positive X-domain; determine a set of fitting parameters for the respective central surfaces and the respective equatorial surfaces based on the transition coordinate and a plurality of constraints; and obtain the profile based on the set of fitting parameters for the respective central surfaces and the respective equatorial surfaces.
2. The system of claim 1, wherein: the controller is configured to select an intraocular lens based at least partially on the profile of the lens capsule.
3. The system of claim 1, wherein transforming the imaging data to the adjusted frame of reference includes: fitting the posterior datapoints and the anterior datapoints to a first circle and a second circle, respectively, and determining intersection points of the first circle and the second circle; and transforming the posterior datapoints and the anterior datapoints such that they are centered about a respective center of the intersection points and rotated such that a tilt angle of rotation is zero.
4. The system of claim 1, wherein: the controller is configured to fit the respective central surfaces in the adjusted frame of reference to respective conic equations for a conic surface; and the

controller is configured to determine a conic x-intercept as a coordinate on the first axis (X) where the respective conic equations intersect in a positive-x domain, the transition coordinate being a product of the conic x-intercept and a predefined constant, the predefined constant being less than 1.

5. The system of claim 1, wherein: the respective central surfaces include a central anterior surface; the controller is configured to represent the central anterior surface as an elliptical cone characterized by a first plurality of variables (Ka, Qa, Ra), the first plurality of variables (Ka, Qa, Ra) being obtained by fitting the imaging data to the central anterior surface in the predefined central region in the adjusted frame of reference; and the central anterior surface Ca(x) is defined

$$\text{as: } Ca(x) = Ka - \frac{\sqrt{-(1 + Qa)[(1 + Qa)^2 x^2 - Ra^2]}}{(1 + Qa)}.$$

6. The system of claim 1, wherein: the respective central surfaces include a central posterior surface; the controller is configured to represent the central posterior surface as an elliptical cone characterized by a second plurality of variables (Kp, Qp, Rp), the second plurality of parameters (Kp, Qp, Rp) being obtained by fitting the imaging data to the central posterior surface in the predefined central region in the adjusted frame of reference; and the central posterior surface Cp(x)

$$\text{is defined as: } Cp(x) = Kp + \frac{\sqrt{-(1 + Qp)[(1 + Qp)^2 x^2 - Rp^2]}}{(1 + Qp)}.$$

7. The system of claim 1, wherein: the set of fitting parameters include a first anterior parameter (Ga), a second anterior parameter (Pa), a first posterior parameter (Gp), a second posterior parameter (Pp) and respective coordinates (Xe, Ye) of a vertex in the adjusted frame of reference; and the respective equatorial surfaces include an equatorial anterior surface and an equatorial posterior surface meeting at the vertex.

8. The system of claim 7, wherein: the equatorial anterior surface and the equatorial posterior surface are represented by respective skewed parabola functions; and the equatorial anterior surface is based in part on the first anterior parameter (Ga), the second anterior parameter (Pa) and the respective coordinates (Xe, Ye) of the vertex, the equatorial anterior surface Ea(x) being defined as:

$$Ea(x) = -[(1 - Ga(Xe - x))][2\sqrt{Pa(Xe - x)} - Ye].$$

9. The system of claim 8, wherein: the equatorial posterior surface is based in part on the first posterior parameter (Gp), the second posterior parameter (Pp) and the respective coordinates (Xe, Ye) of the vertex, the equatorial posterior surface Ep(x) being defined as:

$$Ep(x) = +[(1 - Gp(Xe - x))][2\sqrt{Pp(Xe - x)} + Ye].$$

10. The system of claim 1, wherein: the respective central surfaces include a central posterior surface and a central anterior surface; the respective equatorial surfaces include an equatorial anterior surface and an equatorial posterior surface; and the plurality of constraints includes a first equation matching respective values of the central anterior surface and the equatorial anterior surface at the transition coordinate, and a second equation matching the respective values of the central posterior surface and the equatorial posterior surface at the transition coordinate.

11. The system of claim 10, wherein the plurality of constraints includes: a third equation matching respective first derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate; and a fourth equation matching the respective first derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate.

12. The system of claim 10, wherein the plurality of constraints includes: a fifth equation matching respective second derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate; and a sixth equation matching the respective second derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate.

13. A system comprising: a controller having at least one processor and at least one non-transitory, tangible memory on which instructions are recorded for executing a method for obtaining a profile of a lens capsule of an eye; wherein the profile is represented by respective central surfaces and respective equatorial surfaces separated by respective transition points; wherein execution of the instructions by the processor causes the controller to: obtain a lens diameter and at least two

variables from a set of variables, the set of variables including a lens thickness, a central anterior apex and a central posterior apex; set a transition coordinate as a product of the lens diameter and a predefined constant, the predefined constant being less than 0.5; obtain a first plurality of variables (K_a , Q_a , R_a) and a pair of anterior parameters (G_a , P_a) by simultaneously solving a first group of constraints based in part on the transition coordinate; and obtain a second plurality of variables (K_p , Q_p , R_p) and a pair of posterior parameters (G_p , P_p) by simultaneously solving a second group of constraints based in part on the transition coordinate.

14. The system of claim 13, wherein: the respective central surfaces include a central anterior surface and a central posterior surface, the respective equatorial surfaces including an equatorial anterior surface and an equatorial posterior surface; the controller is configured to represent the central anterior surface and the central posterior surface as respective elliptical cones characterized by the first plurality of variables (K_a , Q_a , R_a) and the second plurality of variables (K_p , Q_p , R_p), respectively; and the controller is configured to represent the equatorial anterior surface and the equatorial posterior surface as respective skewed parabolas characterized by the pair of anterior parameters (G_a , P_a) and the pair of posterior parameters (G_p , P_p), respectively.

15. The system of claim 14, wherein: the controller is configured to select an intraocular lens based at least partially on the profile of the lens capsule.

16. The system of claim 14, wherein the first group of constraints includes: a first equation matching respective values of the central anterior surface and the equatorial anterior surface at the transition coordinate; a second equation matching respective first derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate; a third equation matching respective second derivatives of the central anterior surface and the equatorial anterior surface at the transition coordinate; and a fourth equation matching a respective coordinate of the central anterior surface to the central anterior apex.

17. The system of claim 14, wherein the second group of constraints includes: a fifth equation matching the respective values of the central posterior surface and the equatorial posterior surface at the transition coordinate; a sixth equation matching respective first derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate; a seventh equation matching respective second derivatives of the central posterior surface and the equatorial posterior surface at the transition coordinate; and an eighth equation matching a respective coordinate of the central posterior surface to the central posterior apex.

18. The system of claim 14, wherein the controller is configured to: obtain the profile based on the first plurality of variables (K_a , Q_a , R_a), the pair of anterior parameters (G_a , P_a), the second plurality of variables (K_p , Q_p , R_p) and the pair of posterior parameter (G_p , P_p); obtain an updated value of the lens diameter based on the profile, and an updated value of the transition coordinate based on the updated value of the lens diameter; and when a difference between the updated value of the transition coordinate and the transition coordinate is greater than a predefined threshold, update the first plurality of variables (K_a , Q_a , R_a) and the pair of anterior parameters (G_a , P_a) by simultaneously solving the first group of constraints based in part on the updated value of the transition coordinate; and when the difference between the updated value of the transition coordinate and the transition coordinate is greater than the predefined threshold, update the second plurality of variables (K_p , Q_p , R_p) and the pair of posterior parameter (G_p , P_p) by simultaneously solving the second group of constraints based in part on the updated value of the transition coordinate.

19. A method for obtaining a profile of a lens capsule of an eye with a controller having at least one processor and at least one non-transitory, tangible memory, the method comprising: obtaining imaging data for a portion of the lens capsule visible through a pupil of the eye, the imaging data including posterior datapoints and anterior datapoints; transforming the imaging data to an adjusted frame of reference having a first axis (X) and a second axis (Y), via the controller, the profile of the lens capsule being represented by respective central surfaces and respective equatorial surfaces

separated by respective transition points in the adjusted frame of reference; fitting the imaging data to the respective central surfaces in a predefined central region in the adjusted frame of reference, via the controller; obtaining a transition coordinate as a coordinate value of the respective transition points in a positive X-domain, via the controller; determining a set of fitting parameters for the respective central surfaces and the respective equatorial surfaces based on the transition coordinate and a plurality of constraints, via the controller; and obtaining the profile based on the set of fitting parameters for the respective central surfaces and the respective equatorial surfaces, via the controller.
